SUBHRONIL MUKHOPADHYAY

AI/ML Professional

Technology professional with hands-on experience delivering **Innovative Software Solutions, Data-Driven Systems, and AI Applications** while growing within the IT and Digital Services sector.

Profile Summary

- Emerging AI and software professional with strong hands-on experience in Machine Learning, Data-Driven Systems, and **end-to-end Software Development within IT and Digital Service domains.**
- Contributed to advanced projects at **Hitachi India Pvt. Ltd.**, developing predictive analytics solutions, automated validation frameworks, and interactive visualization tools for industrial technology use cases.
- Proficient in **neural networks, convolutional models, cloud platforms, and scalable software architectures**, with practical exposure through internships and academic projects.
- Strong understanding of **AI Research Methodologies**, Algorithm Optimization, and Collaborative Software Development Practices.
- Displayed effective **communication, leadership, and cross-functional collaboration skills**, enabling seamless coordination between Data Science, Engineering, and Operations Teams.

Work Experience

Jul'26-Present

Sr. Technical Engineer (Contract via Hitachi Systems) | Hitachi India Pvt. Ltd., Bengaluru

Key Result Areas:

- Engineering an **agentic travel-planning middleware** that converts a single natural-language prompt into a complete end-to-end itinerary, autonomously orchestrating train, hotel, and event bookings across 3 Model Context Protocol (MCP) servers exposing 6 tools.
- Architecting a **stateful 6-node LangGraph workflow** with both deterministic and LLM-driven routing across 7 use cases, backed by a dynamic registry that auto-discovers and spawns each MCP server as an isolated subprocess and reflects their tool schemas at runtime.
- Building a **1,000+ line Manim explainer video** for GSK equipment-maintenance SOPs, animating a 5-stage process-discovery pipeline spanning raw demonstration capture, 3D embedding, normalization, and finite-state-machine (FSM) parsing.
- Rendering **interactive 3D scenes** (ThreeDAxes, Dot3D, animated camera orbits) and an FSM that resolves 10 canonical steps into 4 validated task orderings, translating a complex ML pipeline into a stakeholder-ready visual.

Internships

Jan'26 – Jun'26

Research and Development Intern | Hitachi India Pvt. Ltd., Bengaluru

Key Result Areas:

- Designing and implementing an **end-to-end ML-based predictive maintenance system** to enable proactive failure prediction and reduce service disruptions.
- Developing a **rule-based validation framework for GSK machinery maintenance** to improve compliance accuracy and ensure adherence to operational standards.
- Integrating **real-time acoustic signal analysis to validate task completion**, minimizing procedural errors and strengthening operational reliability.
- Building an **interactive dashboard** that showcases critical data center and operational KPIs, enabling faster insights and data-driven decision-making.

Highlights:

- Analyzed large-scale transactional datasets from **500+ ATMs, processing ~10,000 records per ATM monthly** to identify recurring fault patterns and operational anomalies.
- Implemented **automated anomaly detection algorithms**, reducing manual inspection effort and accelerating issue resolution for Maintenance Teams.

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Areas of Exposure

- Cloud Computing Infrastructure
- Predictive Analytics
- Data Visualization and Dashboarding
- Neural Network Architectures
- Real-time Collaborative Systems
- Computer Vision Applications
- Algorithm Optimization
- Software Development Life Cycle

Soft Skills

- Analytical Thinking
- Willingness to Learn
- Innovative Thinking
- Team Player
- Time Management
- Problem-solving

- Collaborated with **Cross-Functional and Cloud Teams** to deploy scalable AWS-based solutions, ensuring secure data handling and efficient model execution.
- Achieved a **40–50% reduction in ATM downtime and enhanced validation accuracy to ~89%** through continuous AI model evaluation, refinement, and stakeholder feedback.

Jun'24 – Jul'24
Machine Learning Intern | BeyonData Solutions Pvt. Ltd., Ahmedabad
Highlights:

- Conducted comprehensive benchmarking of **5 CNN architectures** (VGG16, ResNet50, InceptionV3, MobileNetV2, EfficientNetB0) on a **27-class logo dataset, evaluating performance across multiple metrics.**
- Developed and optimized **data augmentation pipelines** to improve model generalization, achieving a peak classification **accuracy of 93.8% using MobileNetV2**.
- Synthesized experimental results into a **structured comparative framework**, clearly outlining trade-offs between accuracy, inference speed, and computational cost.
- Executed detailed model evaluations across **key performance indicators, including precision, recall, F1-score, inference latency, and model size**, to identify optimal deployment candidates.

Academic Projects

TimeCapsule Connect | MERN Stack, PostgreSQL - July 2025
Highlights:

- Designed and developed a scalable full stack digital time-capsule platform with time- and location-based unlocking, **supporting 100,000+ unique unlock combinations over a 20-year span using KSUIDs and Google Maps API** for precise geolocation.
- Implemented real-time collaborative editing with **SlateJS and Yjs** (<100 ms latency for 10+ users) and **optimized large media uploads** (>1GB) via chunked processing, reducing upload failures **by 60%** and improving user experience.

Dynamic Crowd Simulation | GNN, Transformer, Pytorch, PPO, OpenCV, Python - March 2025
Highlights:

- Designed a hybrid trajectory-prediction model combining Graph Neural Networks and Transformer architectures, **achieving 99%+ accuracy in pedestrian movement forecasting.**
- Integrated a PPO-based reinforcement learning agent for collision-free navigation and trained the system over **20,000+ GPU timesteps**, resulting in a **+6.31 mean reward improvement and strong real-world robustness.**

Advance Voting System | HTML, CSS, JavaScript, NodeJS, OpenCV, TensorFlow, keras, Python - May 2024
Highlights:

- Designed and developed a secure online voting platform with **Digilocker and Voter ID–based authentication**, ensuring robust voter identity verification and data security.
- Enabled remote mobile voting to improve accessibility and participation, while architecting the system to reliably coordinate with **10,000+ daily visitors** and support **1,000+ active users** with consistent performance.

Player Injury Prediction | ANN, TensorFlow, keras, Python - August 2023
Highlights:

- Developed and deployed ANN-based injury prediction models for NBA players, achieving **89% accuracy and a 90% success rate**, forecasting injury types and trends to improve prevention strategies by **25%, reduce injury-related downtime by 40%, and enhance overall team performance by 15%.**

Certifications

- **Gate Qualified** - Gate, Mar 2026
- **Amazon ML Summer School** - Amazon, Sept 2025
- **Microsoft Azure Ai900** - Microsoft, Aug 2024

Personal Details

Date of Birth: 19th December 2004
Languages Known: English, Hindi, Gujarati & Bengali

Technical Skills

- Python, C++, Java, & JavaScript
- AI & Machine Learning Frameworks
- AWS, PostgreSQL, MySQL, & MongoDB
- ReactJS, NodeJS, jQuery
- MS Office Suite
- Git, Figma, OrCAD & Verilog HDL

Education

Complete (Sept'22 – May'26)	B.Tech. Computer Science, Vellore Institute of Technology | Vellore CGPA: 8.88

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